Exact versus Kohn–Sham for a one-dimensional two-electron system

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1 Problem statement

Problem: Benchmark restricted Hartree–Fock against exact solutions of the one-dimensional soft-Coulomb H^-, He and H_2, and measure the correlation energy, including how it grows as H_2 is stretched.

Goal. Solve a two-electron problem on a line exactly. Then solve the same problem with restricted Hartree–Fock, which for two electrons is the same as Kohn–Sham with exact exchange, and compare the energies and densities of the two methods. The difference between the two energies is exactly the correlation energy, so the error of the mean-field method can be measured directly.

Model. We use atomic units throughout: lengths in bohr, energies in hartree (Ha), $\hbar = m_e = e = 1$. Every Coulomb interaction is replaced by the soft-Coulomb form $w(x)$, and the two electrons are described by

$$w(x) = \frac{1}{\sqrt{x^2+1}}, \qquad \hat{H} = -\tfrac{1}{2}\partial_{x_1}^2 - \tfrac{1}{2}\partial_{x_2}^2 + v_{\text{ext}}(x_1) + v_{\text{ext}}(x_2) + w(x_1 - x_2). \tag{1}$$

Table 1 lists the three systems we study.

System	External potential $v_{\text{ext}}(x)$	Note
H^-	$-w(x)$: one nucleus of charge 1 at $x = 0$	Weakly bound
He	$-2w(x)$: one nucleus of charge 2 at $x = 0$	Tightly bound
H_2	$-w(x - R/2) - w(x + R/2)$: two nuclei a distance R apart	Total energy includes $w(R)$

Table 1: The three two-electron systems.

Tasks.

1. **Grid tools.** Build a uniform grid on $[-L, L]$ with spacing h and an odd number of points, the three-point finite-difference kinetic matrix, and a one-electron solver for the lowest eigenpair of $T + \text{diag}(V)$.

2. **Exact solver.** Build the sparse Hamiltonian $T \otimes I + I \otimes T + \text{diag}(V)$ on the (x_1, x_2) grid, find its lowest eigenpair with the Lanczos method, and compute the density $n(x)$ from $\Psi(x_1, x_2)$. Do this for H^-, He and H_2.

3. **Hartree–Fock solver.** Solve for one doubly occupied orbital with a self-consistent loop. Start from the bare-nucleus orbital, build $v_{\text{eff}} = v_{\text{ext}} + \frac{1}{2}v_H$, mix the density with $\alpha = 0.4$, and stop when the change in the density falls below 10^{-6}. For H_2, keep the density symmetric about $x = 0$.

4. **Energy.** Compute $E = T_s + E_{\text{ext}} + E_H + E_x$ term by term, adding $E_{nn} = w(R)$ for H_2, and check it against the eigenvalue route $2\varepsilon - \frac{1}{2}E_H$ (plus E_{nn}).

5. **Comparison and checks.** Compare the exact and Hartree–Fock energies and densities for H^-, He and H_2 at $R = 3.5$. Trace the H_2 energy curve from $R = 1$ to 10 and measure how often both electrons sit on the same side of the molecule. Test convergence in the grid spacing, with Richardson extrapolation, and in the box size, and check how the loop behaves for stretched H_2.

1.1 Phases

The original problem statement divides the work into six phases. This project carries out Phases 1–3. Phases 4–6 are future extensions: we have not started any of them, and Section 9 outlines how they could be done.

Completed.

Phase 1. One-electron solver on a uniform grid with a three-point kinetic energy, and Richardson extrapolation.

Phase 2. Exact two-electron solver: a sparse Hamiltonian on the (x_1, x_2) grid and its lowest eigenstate.

Phase 3. Kohn–Sham with exact exchange, which equals restricted Hartree–Fock for two electrons, giving the exact correlation energy $E_c = E_{\text{exact}} - E_{\text{HF}}$.

Future extensions.

Phase 4. LDA exchange for the 1D soft-Coulomb electron gas.

Phase 5. LDA correlation from a published parametrisation.

Phase 6. The exact exchange–correlation potential, obtained by inverting the exact density.

Our results measure the exact correlation energy and the full error of Hartree–Fock; the error of an approximate density functional such as LDA remains to be measured.

1.2 Code and repository

The code is at `https://github.com/PiCamel/Exact-Against-HF-1D-Soft-Coulomb-Solver` and is laid out as in Table 2. Each solver file has a `run(...)` function that returns a dictionary of results and can also be run on its own (`python KS/KS_He.py`). After installing the three Python packages with `pip install -r requirements.txt`, running `python compare.py` reproduces every number and figure in Section 7 in one to two minutes.

Path	Contents
`common.py`	Grid, w, potentials, kinetic matrix, two-electron Hamiltonian, Hartree potential, one-electron solver, Richardson extrapolation and order of convergence
`exact/`	Exact solvers (Phase 2): `exact_H_minus.py`, `exact_He.py`, `exact_H2.py`
`KS/`	Kohn–Sham = Hartree–Fock (Phase 3): `KS_H_minus.py`, `KS_He.py`, `KS_H2.py`
`compare.py`	Runs every solver, prints all numbers quoted here, saves the plots
`results/`	`energy_errors.png`, `H2_energy_curve.png`, `densities.png`
`report/`	This report
`requirements.txt`	Python packages needed: NumPy, SciPy, Matplotlib

Table 2: Layout of the repository.

2 Why computation is necessary

Limits of analytical methods. Only a handful of potentials, such as the hydrogen atom and the harmonic oscillator, have closed-form solutions, and the soft-Coulomb potential is not one of them: even a single electron in $-Z\,w(x)$ has to be solved numerically. With two electrons the repulsion $w(x_1 - x_2)$ couples the coordinates, so the Schrödinger equation becomes a two-dimensional eigenvalue problem that cannot be separated into one-electron equations. Perturbation theory and variational trial functions give estimates whose error is unknown, so neither can serve as the exact reference this project needs. The softening is itself required: the bare $1/|x|$ interaction is too singular in one dimension to have a finite ground-state energy, and $w(x)$ is the standard choice for one-dimensional model atoms. On a grid the problem becomes linear algebra with an error we can measure and reduce.

Why approximations are still needed. The exact wavefunction depends on the position of every electron at once. With N_g grid points per coordinate in d dimensions,

$$\text{unknowns (exact)} = \left(N_g^d\right)^N, \qquad \text{unknowns (orbitals)} \approx N \cdot N_g^d. \tag{2}$$

For two electrons with $N_g = 50$ in 3D the exact Ψ has 1.6×10^{10} entries (about 125 GB), while one orbital has 125,000. Density functional theory (DFT) avoids this wall by working with the density

$n(x)$, a function of a single position, and gathering the hard physics into one term that has to be approximated [1, 2].

In plain words

Two dice need a 6×6 table of joint probabilities. If the dice ignore each other, two lists of six numbers are enough. With ten dice the joint table has $6^{10} \approx 6 \times 10^7$ entries, against 60 for the lists. The exact wavefunction is the joint table; orbitals are the separate lists.

3 The exact solution on a grid (Phases 1 and 2)

3.1 Symmetry, spin and the density

The Hamiltonian contains no spin. Its ground state has no nodes, so it is symmetric under $x_1 \leftrightarrow x_2$ and pairs with the antisymmetric spin singlet. We therefore solve for one real, positive, symmetric $\Psi(x_1, x_2)$, with density

$$n(x) = 2 \int |\Psi(x, x')|^2 \, \mathrm{d}x', \qquad \int n(x) \, \mathrm{d}x = 2. \tag{3}$$

To elaborate, since electrons are fermions the total wavefunction must be antisymmetric under exchange of the two electrons. Pauli's exclusion principle suggests that in the case of fermions switching the two particles should result in a sign flip of the overall state. Since, the spatial part of the wavefunction is symmetric, the spin part must be antisymmetric to ensure that the total wavefunction is antisymmetric.

3.2 Finite difference method

A function is stored at the grid points $x_i = -L + ih$, and an integral becomes $h \sum_i f_i$. Adding the Taylor expansions of $\psi(x \pm h)$ cancels the odd terms and gives

$$\psi''(x_i) \approx \frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{h^2} \qquad (\text{error } O(h^2)). \tag{4}$$

The operator $-\frac{1}{2}\partial_x^2$ becomes a tridiagonal matrix T (`T_1e`), a potential becomes a diagonal matrix, and the Schrödinger equation becomes the eigenvalue problem $(T+V)\boldsymbol{\psi} = E\boldsymbol{\psi}$. Its lowest eigenvector is the ground state (`one_electron`).

3.3 Two electrons: the Kronecker product

$\Psi(x_1, x_2)$ is an $N \times N$ table, flattened into a vector of N^2 entries. Each electron's kinetic energy acts along its own index:

$$H_{2e} = T \otimes I + I \otimes T + \mathrm{diag}(V), \qquad V_{ij} = v(x_i) + v(x_j) + w(x_i - x_j). \tag{5}$$

For $N = 201$ this matrix has 40,401 rows with about 5 non-zeros each. We store it sparse (`H_two_electron`) and find the lowest eigenpair with the Lanczos method (`eigsh`). Because physics needs $h \sum \psi_i^2 = 1$, the eigenvector is divided by $\sqrt{h}$ in 1D and by h in 2D.

3.4 Richardson extrapolation

If the error is ch^2, then $E(h) = E_0 + ch^2$ and $E(2h) = E_0 + 4ch^2$, and eliminating c gives

$$E_0 \approx E(h) + \frac{E(h) - E(2h)}{3}. \tag{6}$$

Three spacings $h, 2h, 4h$ give the observed order $p = \log_2[(E_{2h} - E_{4h})/(E_h - E_{2h})]$, which should be 2. Both are implemented in `common.py` (`richardson`, `convergence_order`).

Insight from the project

Check $h\sum_i n_i = 2$ and the h^2 scaling. Every solver gives $\int n = 2.0$, and the helium energy differences shrink by factors 4.07 and 4.02 as h is halved (Table 5). A missing factor h in a sum that stands in for an integral is the most common silent bug.

4 Density functional theory: Hohenberg–Kohn and Kohn–Sham

Hohenberg–Kohn. The ground-state density fixes the external potential and hence every property of the system [1]. The energy is a functional of the density, $E[n] = F[n] + \int n\, v_{\text{ext}}\, \mathrm{d}x \geq E_0$, and the true density minimises it. The difficulty is that the kinetic part of $F[n]$ is unknown.

Kohn–Sham. Replace the interacting electrons by non-interacting ones with the *same density*. Their kinetic energy T_s is easy to compute from orbitals, and everything else goes into the exchange–correlation energy [1, 2, 3]:

$$E[n] = T_s[n] + \int n\, v_{\text{ext}}\, \mathrm{d}x + E_H[n] + E_{xc}[n], \qquad \Big[-\tfrac{1}{2}\partial_x^2 + v_{\text{ext}} + v_H + v_{xc}\Big]\varphi_i = \varepsilon_i \varphi_i, \tag{7}$$

with $v_{xc} = \delta E_{xc}/\delta n$. With the exact E_{xc} the method would be exact, so every practical error comes from the approximation chosen for it. Section 5.2 shows which one we use.

Note

The Kohn–Sham electrons are not the true electrons. They do not interact, but on computing density, one cannot tell the difference at first glance. The exchange–correlation potential v_{xc} accounts for the difference between these fictitious non-interacting electrons and the real interacting electrons. The exact v_{xc} is unknown, so approximations are used in practice. In our case, v_{xc} is taken to be $-\frac{1}{2}v_H$, the exact exchange potential for two electrons in one orbital; correlation is left out.

5 Hartree–Fock and the self-consistent loop (Phase 3)

5.1 One orbital for both electrons

The simplest wavefunction puts both electrons (opposite spins) in one orbital, $\Psi = \varphi(x_1)\varphi(x_2)$, with $n = 2\varphi^2$. The Hartree potential and energy of the electron cloud are

$$v_H(x) = \int n(x')\, w(x - x')\, \mathrm{d}x', \qquad E_H = \tfrac{1}{2}\int n\, v_H\, \mathrm{d}x = \tfrac{1}{2}\iint n(x)\, n(x')\, w(x - x')\, \mathrm{d}x\, \mathrm{d}x'. \tag{8}$$

On the grid, $v_H = h\, W n$ with $W_{ij} = w(x_i - x_j)$.

The $\frac{1}{2}$ in E_H prevents double counting. The density holds the charge of both electrons, and the double integral meets every pair of points twice, once as (x, x') and again as (x', x). Halving the integral counts each repulsion once. This factor is needed for any number of electrons and is separate from the one in Section 5.2, which is special to two.

5.2 Equivalence of HF and KS for this problem

Since v_H is built from the whole cloud, each electron also repels its own charge. For two electrons in one orbital each electron is exactly half the cloud, so exact exchange removes exactly half:

$$E_x = -\tfrac{1}{2}E_H, \qquad v_x = -\tfrac{1}{2}v_H, \qquad v_{\text{eff}} = v_{\text{ext}} + \tfrac{1}{2}v_H. \tag{9}$$

Hartree–Fock and Kohn–Sham both arrive at this equation:

- *Hartree–Fock.* The orbital equation is $\big[-\frac{1}{2}\partial_x^2 + v_{\text{ext}} + 2J - K\big]\varphi = \varepsilon\varphi$, where J multiplies by $\frac{1}{2}v_H$. With a single orbital the exchange operator gives $K\varphi = J\varphi$, so $2J - K$ reduces to $\frac{1}{2}v_H$.

- *Kohn–Sham.* With $E_{xc} = -\frac{1}{2}E_H$ and no correlation, $v_{xc} = -\frac{1}{2}v_H$, so $v_{\text{ext}} + v_H + v_{xc} = v_{\text{ext}} + \frac{1}{2}v_H$.

The orbital, density and energy are therefore identical. Our `KS/` solvers are Hartree–Fock solvers, and we use "HF", "RHF" and "KS" for the same calculation. With more than two electrons the two methods differ, because Hartree–Fock exchange becomes a non-local operator.

5.3 The energy, term by term

$$E = \underbrace{2\langle\varphi| -\tfrac{1}{2}\partial_x^2|\varphi\rangle}_{T_s} + \underbrace{\int n\, v_{\text{ext}}\,\mathrm{d}x}_{E_{\text{ext}}} + \underbrace{\tfrac{1}{2}\int n\, v_H\,\mathrm{d}x}_{E_H} + \underbrace{(-\tfrac{1}{2}E_H)}_{E_x}\ (+\ E_{nn}). \tag{10}$$

T_s is the cost of confinement, E_{ext} the attraction to the nuclei, E_H the cloud repelling itself and E_x the removal of the self-repulsion. The correlation energy $E_c = E_{\text{exact}} - E_{\text{HF}} \le 0$ is what a product state misses, because real electrons avoid each other.

Doubling the orbital equation gives $2\varepsilon = T_s + E_{\text{ext}} + E_H$, so $E = 2\varepsilon - \frac{1}{2}E_H$. Each ε already includes the repulsion from the other electron, and adding both counts that pair twice. Our code computes E both ways as a check.

Nuclear repulsion. For H_2 both solvers add $E_{nn} = w(R)$. `exact_H2.py` returns `E[0] + E_nn` and keeps the electronic energy as `E_elec`; `KS_H2.py` adds `E_nn_H2` in both energy routes.

- *Without E_{nn}* the number is the *electronic energy*: the energy of two electrons around clamped nuclei.
- *With E_{nn}* it is the *total energy of the molecule* at that geometry.

At fixed R, E_{nn} is a constant. It shifts every eigenvalue by the same amount, leaves Ψ and n unchanged, and cancels in E_c. Between geometries it matters: the electronic energy keeps falling as the nuclei approach (from -1.439 Ha at $R = 10$ to -2.120 Ha at $R = 1$, Table 6), so on its own it would predict no bond. Only the total energy has a minimum. For a single atom $E_{nn} = 0$.

5.4 The self-consistent loop

The potential depends on n and n depends on the potential, so we iterate (Figure 1). Jumping straight to the new density can overshoot, so we mix:

$$n_{\text{in}}^{(k+1)} = \alpha\, n_{\text{out}}^{(k)} + (1-\alpha)\, n_{\text{in}}^{(k)}, \qquad \text{stop when } \sqrt{\textstyle\sum_i (n_{\text{out},i} - n_{\text{in},i})^2} < 10^{-6}. \tag{11}$$

We use $\alpha = 0.4$ in every calculation.[1] The convergence measure has no grid weight h, so the tolerance means slightly different things on different grids; this has no visible effect at the precision quoted.

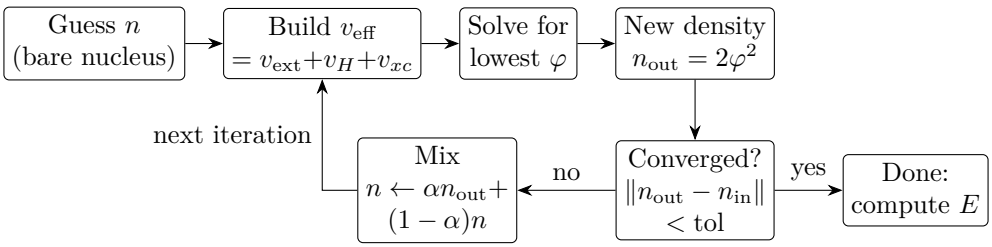


Figure 1: The self-consistent field loop. Here $v_{xc} = -\frac{1}{2}v_H$, which makes it the Hartree–Fock loop.

[1] The mixing parameter controls how fast the density settles. A larger α takes bigger steps and usually needs fewer iterations, as long as the update does not overshoot; when it does, the error oscillates and can grow. A smaller α damps the overshoot at the price of more iterations. Near the solution the error shrinks by about $|1 - \alpha + \alpha\lambda|$ per iteration, where λ measures how the output density responds to the input density, so convergence needs this factor below 1. We did not vary α in this project.

Insight from the project

Stretched H_2 needs a symmetric density. At $R = 6$ the plain loop never settles: the charge sloshes from one nucleus to the other and the error is still 2.67 after 300 iterations. Imposing the mirror symmetry of the exact ground state, $n(x) = n(-x)$, at every step makes it converge in 22 iterations. All H_2 results use the symmetrised loop.

6 Stretched H_2: static correlation

With two nuclei at $\pm R/2$, the exact energy tends to that of two separate hydrogen atoms as $R \to \infty$, $2E(\mathrm{H}) = -1.33972\,\mathrm{Ha}$ on our grid (output 2a). Restricted Hartree–Fock keeps one orbital $\varphi = (\chi_A + \chi_B)/\sqrt{2}$ for both electrons, and the product expands into

$$\Psi_{\mathrm{RHF}} = \tfrac{1}{2}[\chi_A(x_1)\chi_A(x_2) + \chi_B(x_1)\chi_B(x_2) + \chi_A(x_1)\chi_B(x_2) + \chi_B(x_1)\chi_A(x_2)]. \tag{12}$$

The last two terms describe neutral $\mathrm{H} + \mathrm{H}$, the first two put both electrons on one atom ($\mathrm{H}^- + \mathrm{H}^+$). RHF is therefore half ionic at every distance. The true state at large R is purely covalent, which one shared orbital cannot describe; this failure is called static correlation. Spin-unrestricted HF would fix the energy at the cost of spin symmetry; we did not run it.

In plain words

Two people in two far-apart rooms. In reality there is one person per room. Restricted HF places each person independently "half in each room", so half the time both end up in the same room.

7 Results

Every number below is printed by `compare.py`; the tag in each caption (for example "output 1a") names the block of its output. Unless stated otherwise, $h = 0.1$ and $L = 10$.

7.1 Helium and the hydrogen anion

After Richardson extrapolation the solvers reproduce the problem statement's reference values to four decimals (output 3b): one-electron H $= -0.6698$, $\mathrm{He}^+ = -1.4834$, exact He $= -2.2383$, RHF $= -2.2242$ and $E_c = -0.0140\,\mathrm{Ha}$.

System	E_{exact}	E_{RHF}	$\Delta_{\mathrm{RHF}} = -E_c$	δ_{RHF} (%)
He, $h = 0.1$	-2.23855	-2.22450	0.01405	0.63
He, extrapolated	-2.23826	-2.22421	0.01405	0.63
H^-, $L = 10$	-0.73043	-0.69182	0.03861	5.29
H^-, $L = 14$	-0.73079	-0.69208	0.03871	5.30

Table 3: Total energies in hartree (outputs 1a, 3b, 4). $\Delta = E_{\mathrm{RHF}} - E_{\mathrm{exact}}$, $\delta = 100|\Delta|/|E_{\mathrm{exact}}|$.

System	T_s	E_{ext}	E_H	E_x	E_{nn}	E (terms)	E (2ε route)	iter.
H^-	0.1131	-1.3920	1.1743	-0.5871	0	-0.69182	-0.69182	23
He	0.2905	-3.2389	1.4477	-0.7239	0	-2.22450	-2.22450	23
H_2 ($R = 3.5$)	0.0965	-2.1652	1.0595	-0.5298	0.2747	-1.26423	-1.26423	19

Table 4: Hartree–Fock energy terms in hartree (output 1b). The two routes to E agree to within $6 \times 10^{-8}\,\mathrm{Ha}$.

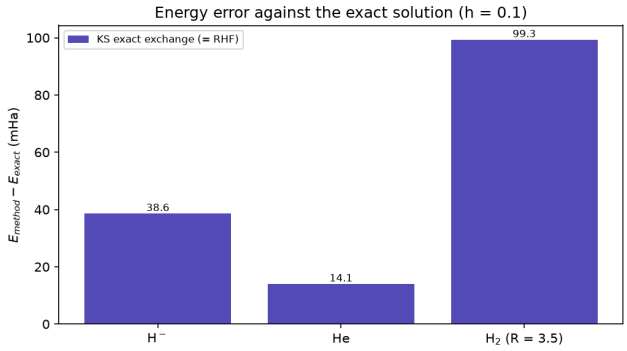


Figure 2: Hartree–Fock energy error $E_{\mathrm{RHF}} - E_{\mathrm{exact}} = -E_c$ in millihartree; H_2 at $R = 3.5$.

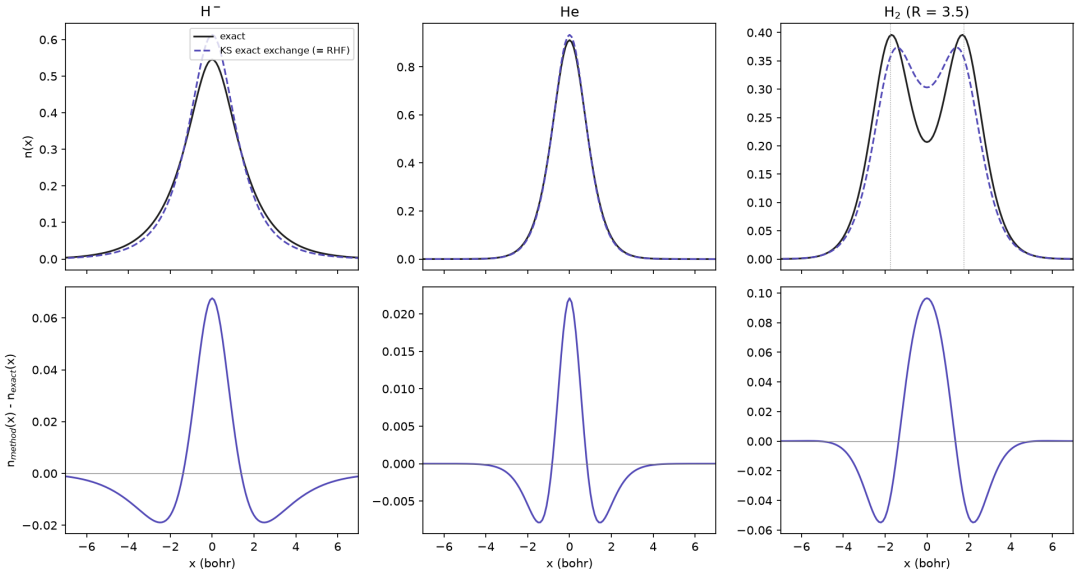


Figure 3: Top: densities, exact (solid) and Hartree–Fock (dashed). Bottom: $n_{\mathrm{RHF}} - n_{\mathrm{exact}}$. Dotted lines mark the H_2 nuclei.

Insight from the project

Correlation is small in helium and large in the anion. The correlation energy of He is only 0.014 Ha (0.63% of the total). In H^- it is 0.039 Ha, almost three times larger, because the loosely bound electrons have room to avoid each other and a product wavefunction cannot describe that. The densities show the same pattern (output 1c): the He peaks nearly coincide (0.933 against 0.911), while the RHF density of H^- is too compact (0.613 against 0.545). H^- is also so diffuse that its exact energy still changes in the fourth decimal place between $L = 10$ and $L = 14$ (Table 5), whereas helium is converged to 10^{-6} Ha by $L = 8$.

7.2 Grid and box convergence

h	E_{exact}(He)	ratio	E_{RHF}(He)	ratio
0.4	−2.243017	–	−2.228938	–
0.2	−2.239430	–	−2.225374	–
0.1	−2.238550	4.07	−2.224500	4.08
0.05	−2.238331	4.02	−2.224282	4.02
extrap.	−2.238258	–	−2.224209	–

L	E_{exact}(He)	$E_{\mathrm{exact}}(H^-)$	$E_{\mathrm{RHF}}(H^-)$
6	−2.238506	−0.723313	−0.686460
8	−2.238549	−0.729191	−0.690923
10	−2.238550	−0.730432	−0.691823
12	−2.238550	−0.730721	−0.692031
14	−2.238550	−0.730790	−0.692082

Table 5: Left: grid convergence of He at $L = 10$ (output 3a); a ratio of successive differences near 4 means error $\propto h^2$. Right: box-size convergence at $h = 0.1$ (output 4). Energies in hartree.

The observed order of convergence is $p = 2.03$ and 2.01 for both solvers, as expected for a three-point second derivative.

7.3 Stretched H_2

R	E_{exact}	E_{RHF}	$\Delta_{\text{RHF}} = -E_c$	δ_{RHF} (%)	$E_{nn} = w(R)$	exact electronic
1.0	−1.41320	−1.39452	0.01868	1.32	0.70711	−2.12030
1.5	−1.45172	−1.42639	0.02533	1.74	0.55470	−2.00642
2.0	−1.44200	−1.40591	0.03609	2.50	0.44721	−1.88921
2.5	−1.41356	−1.36158	0.05198	3.68	0.37139	−1.78495
3.0	−1.38483	−1.31146	0.07338	5.30	0.31623	−1.70106
4.0	−1.35067	−1.22322	0.12745	9.44	0.24254	−1.59321
5.0	−1.34091	−1.16115	0.17976	13.41	0.19612	−1.53703
6.0	−1.33930	−1.12054	0.21876	16.33	0.16440	−1.50370
8.0	−1.33946	−1.07697	0.26249	19.60	0.12403	−1.46350
10.0	−1.33963	−1.05706	0.28257	21.09	0.09950	−1.43913

Table 6: Stretched H_2 at $h = 0.1$, $L = \max(10, R/2 + 9)$, in hartree (output 2a). E_{exact} and E_{RHF} include E_{nn}; the last column is the exact electronic energy alone. Mean absolute RHF error: 0.1276 Ha.

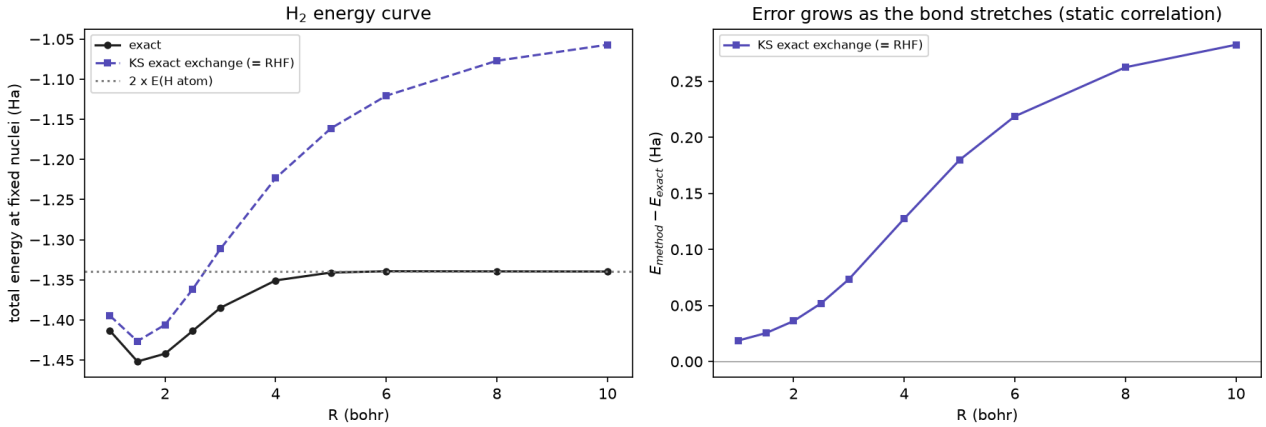


Figure 4: Left: H_2 total energy against R; the grey line is the exact dissociation limit −1.33972 Ha. Right: the RHF error, equal to $|E_c|$.

Insight from the project

Stretching the bond breaks Hartree–Fock. The RHF error grows from 0.019 Ha at $R = 1$ to 0.283 Ha at $R = 10$ and is still rising, while the exact curve has its minimum near $R \approx 1.5$–2 and levels off at the two-atom limit. The wavefunctions show why (output 2b): at $R = 3.5$ both electrons sit on the same side of the molecule with probability 0.145 in the exact Ψ and 0.485 in the RHF product, close to the one half that Section 6 predicts.

Comparison with the problem statement's table. Our exact H_2 energies agree with the reference table at $R = 1, 2, 4, 6, 10$ to about 2×10^{-4} Ha, consistent with its coarser grid ($h = 0.15$). The RHF values agree at $R = 1$ and 2 but differ beyond: at $R = 4$, 6 and 10 the reference lists −1.1904, −1.0693 and −1.0224 Ha, against our −1.2232, −1.1205 and −1.0571 Ha. Since RHF is variational, the lower values belong to a better restricted state. Our loop fails to converge at large R when the density is not symmetrised, so we think the reference values came from such an unconverged run, although we cannot confirm how that table was made.

Insight from the project

Check every number two ways. Computing each energy by two routes and checking $E_{\mathrm{RHF}} \geq E_{\mathrm{exact}}$ exposed the disagreement above. The same habit caught the bugs in our first notebook: two softening lengths in one Hamiltonian (0.2 for the H_2 electron repulsion, 0.001 in one H^- run), a grid spacing hard-coded apart from the grid, the Kohn–Sham loop using the exact solver's grid, and a missing nuclear repulsion.

8 Error analysis

For a method with energy E_{approx} on the same grid as the exact energy,

$$\Delta = E_{\mathrm{approx}} - E_{\mathrm{exact}}, \qquad \delta = \frac{|\Delta|}{|E_{\mathrm{exact}}|} \times 100\%, \qquad \mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |\Delta_i|. \tag{13}$$

Hartree–Fock is variational, so $\Delta_{\mathrm{RHF}} \geq 0$ must hold on every grid; all our values satisfy it. By definition $\Delta_{\mathrm{RHF}} = -E_c = |E_c|$, so the RHF error is the correlation energy itself. Over the ten H_2 distances, $\mathrm{MAE}_{\mathrm{RHF}} = 0.1276\,\mathrm{Ha}$.

Discretisation error (output 3a). The error of $E(h)$ is about $|E(h) - E(2h)|/3$, which is 0.29 mHa for helium at $h = 0.1$. Extrapolating from $(0.2, 0.1)$ and from $(0.1, 0.05)$ gives -2.238256 and $-2.238258\,\mathrm{Ha}$, so the extrapolated energy is good to about $10^{-6}\,\mathrm{Ha}$. The H_2 energies are not extrapolated and should carry a grid error of a few $10^{-4}\,\mathrm{Ha}$.

Propagation. With independent uncertainties, $\sigma_{E_c} = \sqrt{\sigma_{\mathrm{exact}}^2 + \sigma_{\mathrm{HF}}^2}$, and in the worst case $\sigma_{E_c} \leq \sigma_{\mathrm{exact}} + \sigma_{\mathrm{HF}}$. Both solvers share the grid, so much of the h^2 error cancels: E_c is $-0.01405\,\mathrm{Ha}$ both at $h = 0.1$ and after extrapolation (output 3b). Every error Δ we quote is far larger than these uncertainties, apart from the box-limited H^-.

Strengths and limitations of the method.

- *Strengths.* The only approximation in the exact solver is the grid, whose error falls as h^2, is largely removed by Richardson extrapolation, and can be checked by refining h and L. Sharing one grid makes E_c more accurate than either energy, each energy is checked two ways, and the sparse Lanczos solve is fast for two electrons.
- *Limitations.* The exact solve stores $(N_g^d)^N$ numbers and cannot reach more electrons or three dimensions (Section 2). The stencil is only second order, the finite box limits the diffuse H^-, our Hartree potential rebuilds a dense $N_g \times N_g$ matrix every iteration, and the plain mixing loop needs the symmetry fix for stretched H_2. Absolute numbers from the softened 1D model do not carry over to real atoms, but the static-correlation failure it exposes is the one seen in real molecules.

9 Future direction

LDA exchange (Phase 4). LDA treats every slice of the system as a piece of uniform electron gas at the local density [2, 3]. For the 1D soft-Coulomb gas the problem statement gives the exchange energy per unit length. It follows from putting the uniform-gas density matrix $\gamma(x) = (k_F/\pi)\sin(k_F x)/(k_F x)$, with $k_F = \pi n/2$, into the exchange integral and substituting $t = k_F x$:

$$n\,e_x(n) = -\int_{-\infty}^{\infty} w(x)\,\gamma(x)^2\,\mathrm{d}x = -\frac{n^2}{2} J\!\left(\frac{\pi n}{2}\right), \qquad J(k) = \int_0^{\infty} \frac{\sin^2 t}{t^2\sqrt{t^2+k^2}}\,\mathrm{d}t, \tag{14}$$

with $v_x = \mathrm{d}[n\,e_x]/\,\mathrm{d}n$ and $e_x \to -\frac{1}{2}$ at high density.

To implement it, J would be tabulated once and splined in $\ln n$, and one line of our loop would change, to $v_{\mathrm{eff}} = v_{\mathrm{ext}} + v_H + v_x(n)$. The problem statement's reference value for 1D helium is $-2.158\,\mathrm{Ha}$, with

an LDA exchange energy of -0.654 Ha against -0.724 Ha for exact exchange. That shortfall is several times our measured correlation energy.

In plain words

LDA values hilly farmland by treating each small patch as flat land at that patch's altitude. This works where the land changes slowly and fails near cliffs, and atoms are full of cliffs.

Exact v_{xc} (Phase 6). For two electrons the Kohn–Sham orbital is $\varphi = \sqrt{n/2}$, so our exact density gives $v_{xc} = \varepsilon + \varphi''/(2\varphi) - v_{\text{ext}} - v_H$ directly, with the constant ε chosen so that v_{xc} vanishes far from the atom. Its tail should follow $-w(x)$, and for stretched H_2 it should grow a peak at the bond midpoint that no local functional can reproduce.

Other steps. LDA correlation (Phase 5), the Z scan (the one-nucleus solvers already take `Z`), the softening scan, unrestricted Hartree–Fock for stretched H_2, and timing the solvers against grid size.

10 Conclusions

1. The exact solver works for H^-, He and H_2, reproduces the reference values for the atoms, and is accurate to about 10^{-6} Ha for extrapolated helium.
2. Hartree–Fock, identical here to Kohn–Sham with exact exchange, is implemented for the same systems. Two energy routes agree, and $E_{\text{RHF}} \geq E_{\text{exact}}$ holds everywhere.
3. The exact correlation energy is -0.014 Ha for He and -0.039 Ha for H^-, and reaches -0.28 Ha for stretched H_2, where restricted Hartree–Fock fails because of static correlation.
4. Grid and box convergence and the charge-sloshing fix are all reproducible from `compare.py`.

Phases 4–6 (LDA exchange, LDA correlation and the exact exchange–correlation potential) are future extensions of the same problem. The advantages of DFT are made more obvious in systems containing more electrons. The question of where approximate DFT fails is therefore answered here only for exact exchange with no correlation; Section 9 outlines the next steps.

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